

UNISOLE AI CAMPUS PROGRAM

GROUP 2 — SCIENCE & MATHEMATICS

Pathway 2: Mathematics + AI / Computational Intelligence

Target students: Mathematics • Applied Mathematics • Statistics • other strongly quantitative disciplines

Positioning: The more mathematics-oriented science pathway — for students who want depth in the theory behind AI rather than domain-science applications.

Overview

This pathway is structured for mathematics-heavy students who want to understand AI at the level of the underlying algorithms and optimization theory, not just as an applied tool. It moves systematically from pure mathematical foundations through to applied scientific ML, ending in a capstone that is explicitly framed as a mathematical/computational problem.

Why this pathway matters: Many AI courses teach tools without teaching why they work. Mathematics students are equipped to go deeper — understanding the actual optimization theory, probability theory, and numerical methods that make modern AI possible, not just how to call a library function.

Module 1 — Mathematics Behind AI

The formal theoretical foundation for everything that follows — treating AI not as a black box, but as an application of mathematics students already study.

Topics covered:

- Formal mathematical structures underlying learning algorithms
- The connection between abstract mathematics and concrete model behaviour
- Reading and understanding the mathematical notation used in AI research and papers

Skills gained: The ability to read AI research and technical documentation with genuine mathematical understanding, not just intuition.

Module 2 — Optimization

How models actually "learn" — by minimizing error through mathematical optimization. This module goes deeper into the theory than a typical applied ML course would.

Topics covered:

- Optimization theory and convex/non-convex problems
- Gradient-based optimization methods
- How optimization algorithms are chosen and tuned for different problems

Skills gained: A rigorous understanding of the mathematics that actually drives model training, not just a black-box "the model learns" explanation.

Module 3 — Probability & Statistics

The language of uncertainty that underlies every model's predictions — treated with the depth a mathematics student is capable of.

Topics covered:

- Probability theory as applied to model behaviour and predictions
- Statistical inference and its role in evaluating models
- Uncertainty quantification in AI systems

Skills gained: The ability to reason precisely about how confident a model's predictions actually are — a frequently misunderstood but critical skill.

Module 4 — Numerical Computing

Solving mathematical problems computationally at scale — the bridge between pure theory and working code.

Topics covered:

- Numerical methods for problems without closed-form solutions
- Computational efficiency and numerical stability
- Practical implementation of mathematical algorithms

Skills gained: The ability to translate mathematical theory into working, efficient computational code.

Module 5 — Machine Learning

Applying the mathematical toolkit built so far to real learning problems.

Topics covered:

- Core ML algorithms, understood from their mathematical foundations up
- Model behaviour explained through the underlying optimization and probability theory
- Practical application to real datasets

Skills gained: A mathematically grounded understanding of ML algorithms that goes beyond "which function to call."

Module 6 — Neural Networks

The mathematics of deep learning architectures — how layers, weights, and activation functions combine to approximate complex functions.

Topics covered:

- The mathematical structure of neural networks
- How backpropagation works as an application of calculus
- Why certain architectures succeed or fail mathematically

Skills gained: A genuine understanding of why neural networks work, not just how to build one.

Module 7 — Scientific ML

Combining mathematical modelling with data-driven learning — bridging pure mathematics with applied scientific computing.

Topics covered:

- Hybrid mathematical/data-driven modelling approaches
- Applying rigorous mathematics to real prediction problems
- Connecting this pathway's theoretical depth to practical scientific applications

Skills gained: The ability to move fluidly between theoretical mathematics and applied computational modelling.

Module 8 — Optimization-Driven AI

Advanced optimization techniques used in modern AI systems — the most technically demanding module in the pathway.

Topics covered:

- Advanced optimization algorithms used in state-of-the-art AI systems
- The mathematical trade-offs behind modern training techniques
- How cutting-edge AI research pushes optimization theory forward

Skills gained: Exposure to genuinely advanced, research-level mathematical concepts in AI — positioning students strongly for further postgraduate study.

Capstone Project

Students solve a mathematical/scientific problem using computational methods and AI, producing a fully worked, defensible solution — including the mathematical reasoning behind every major choice made, not just the final result.

Tools & technologies used across this pathway: Python, NumPy, SciPy, optimization libraries, numerical computing tools, neural network frameworks.

Career relevance: Quantitative Analyst, Research Mathematician, AI Research roles, Data Scientist (theory-leaning), strong preparation for postgraduate study in mathematics, statistics, or AI research.

Student outputs: A mathematically rigorous computational project + technical report + presentation, suitable as evidence for both industry roles and postgraduate applications.

Industry-Readiness Layer *(applies alongside this pathway)*

Professional Digital Profile: Professional email, LinkedIn, Resume, Portfolio, Project documentation, Professional communication.

Evidence: Git, GitHub, README, Repository structure, Version control for computational work; Portfolio, Research documentation, and Project presentation for the theoretical side.

Interview Preparation: Resume walkthrough, Tell-me-about-yourself, Project explanation, Behavioural questions, Technical/domain questions, Mock interview, Feedback.

Industry Interaction: Weekly/periodic expert sessions, Career sessions, Industry case studies, How companies recruit, How to approach professionals, Networking etiquette.

Project Defence: Every student must be able to answer — What problem did you solve? Why did you choose this approach? What did you build? What did you learn? What are the limitations? What would you improve?

Final Capstone Framework

1. **Problem** — Identify a real-world problem
2. **Research** — Understand the domain and existing solutions
3. **Data/Information** — Collect and organize relevant information
4. **Build** — Create the appropriate mathematical/computational solution
5. **Validate** — Test the solution and interpret results
6. **Document** — Report, Repository/portfolio, Documentation, Presentation
7. **Defend** — Present the project before an evaluator/mentor

What This Student Should Have at the End

Scientific/Computational Project + Notebook/Code + Research Report + Presentation

Training and assessment are program components. Internship, advanced projects, hackathons, research engagement and other opportunities are performance/availability-based and subject to applicable selection procedures.